

# Task – 4

## Build the Basic Research Agent Workflow

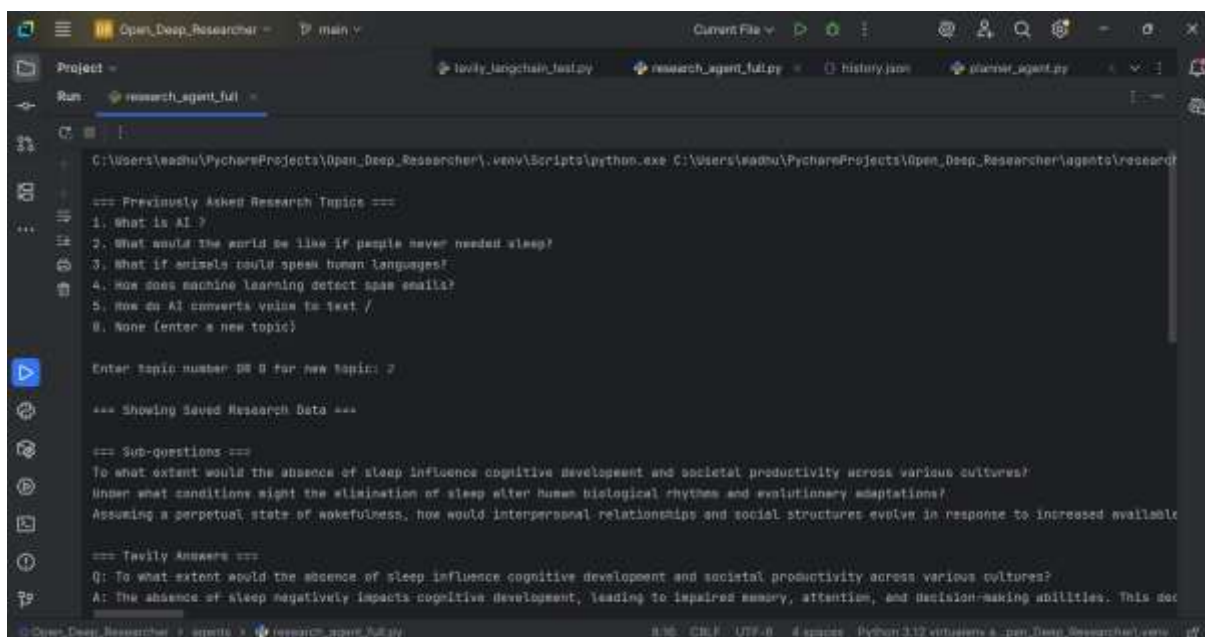
In this project, I built an AI Research Assistant that can take any topic, generate smart sub-questions, search the web using Tavily, find a research paper, summarize the PDF, and give a final combined answer. I also added a history system so I can revisit past topics easily. While doing this, I learned how to integrate LLMs, handle API limits, use chunk-based summarization, and build a reliable end-to-end workflow.

## Advantages

1. Fully automated research flow
2. Works even with free OpenAI API
3. Stores and retrieves past research
4. Handles both scientific and hypothetical topics
5. Intelligent and high-quality sub-questions
6. Reliable PDF summarization

## Limitations

1. Slow execution due to rate-limit delays
2. PDF text extraction sometimes depends on paper quality
3. Heavy LLM usage
4. Large PDFs take longer to summarize



```
C:\Users\madhu\PycharmProjects\Open_Deep_Researcher\.venv\Scripts\python.exe C:\Users\madhu\PycharmProjects\Open_Deep_Researcher\agents\researcher.py

=== Previously Asked Research Topics ===
1. What is AI ?
2. What would the world be like if people never needed sleep?
3. What if animals could speak human languages?
4. How does machine learning detect spam emails?
5. How do AI converts voice to text ?
6. None (enter a new topic)

Enter topic number OR 0 for new topic: 2

=== Showing Saved Research Data ===

=== Sub-questions ===
To what extent would the absence of sleep influence cognitive development and societal productivity across various cultures?
Under what conditions might the elimination of sleep alter human biological rhythms and evolutionary adaptations?
Assuming a perpetual state of wakefulness, how would interpersonal relationships and social structures evolve in response to increased available

=== Tavily Answers ===
Q: To what extent would the absence of sleep influence cognitive development and societal productivity across various cultures?
A: The absence of sleep negatively impacts cognitive development, leading to impaired memory, attention, and decision-making abilities. This not
```

```
Open_Deep_Researcher - main - Current File - tavily_jangchain_test.py - research_agent_full.py - history.json - planner_agent.py -  
Project -  
Run - research_agent_full -  
+++ Tavily Answers +++  
Q: To what extent would the absence of sleep influence cognitive development and societal productivity across various cultures?  
A: The absence of sleep negatively impacts cognitive development, leading to impaired memory, attention, and decision-making abilities. This de  
Q: Under what conditions might the elimination of sleep alter human biological rhythms and evolutionary adaptations?  
A: The elimination of sleep could alter human biological rhythms and evolutionary adaptations by disrupting circadian rhythms, affecting h  
Q: Assuming a perpetual state of wakefulness, how would interpersonal relationships and social structures evolve in response to increased avail  
A: In a perpetual state of wakefulness, interpersonal relationships may evolve to become more intense and frequent due to the increased avail  
+++ Research Paper Summary +++  
**Proble Addressed:**  
The research paper discusses the widespread issue of sleep deprivation, highlighting its prevalence among various age groups in the U.S. and its  
**Methods:**  
The paper synthesizes data from various studies and expert recommendations, including statistics on sleep duration across different age groups.  
**Key Findings:**  
- 35% of U.S. adults sleep less than 7 hours a night; 75% of high school students and 58% of middle school students do not meet recommended slee  
- Common causes of sleep deprivation include poor sleep habits, circadian rhythm disturbances, sleep disorders, and physiological changes in chi  
- Negative effects of sleep deprivation include daytime sleepiness, mood changes, cognitive impairments, chronic fatigue, and increased risk of  
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**Limitations:**  
The paper primarily serves as an educational resource and does not present original research data. It relies on existing studies and expert opir  
**Conclusion:**  
Sleep deprivation is a common and serious issue that can significantly impact health and daily functioning. Prioritizing sleep, adopting good sl  
+++ Final Summary +++  
The research highlights the critical role of sleep in cognitive development and societal productivity, emphasizing that sleep deprivation adve  
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